Lab 5 — Assignment 1 Report: RDS Migration for the Lab 4 Matomo App

Course: Cloud Computing, T.E. Computer, Sem V (2026-2027) | **Account**: calista-labs (386347568553) | **Region**: ap-south-1

EC2: matomo-lab4 (i-077933894a1f47c75), http://15.252.100.78/ | **RDS**: matomo-rds | **Date**: 2026-08-22

Architecture

Browser -> HTTP:80 -> EC2 (matomo-lab4: Nginx + PHP-FPM 8.5 + Matomo) -> TLS, MySQL protocol, port 3306 -> RDS matomo-rds (MariaDB 11.8.8). Security group `matomo-rds-sg` restricts inbound 3306 to the EC2 instance's security group only — no 0.0.0.0/0.

Lab 4's app runs MariaDB (SQL) — Case A of the brief. Deployed engine is **MariaDB 11.8.8**, not generic MySQL, because Matomo's dump uses the MariaDB-only `utf8mb4_uca1400_ai_ci` collation (no MySQL equivalent). MariaDB is literally the same DB type as Lab 4 and is a first-class RDS engine; the restore completed with zero errors, confirming this was correct.

Database & security settings

Setting	Value
Engine / class	MariaDB 11.8.8, db.t3.micro (free-tier)
Deployment	Single-AZ, 20 GiB gp3, autoscaling off
Endpoint	matomo-rds.chu4e4kcinv9.ap-south-1.rds.amazonaws.com:3306
Public access	No
Inbound rule	3306 from EC2 security group (sg-06e7da1b08cdd44a4) only
App DB user	matomo_app — SELECT/INSERT/UPDATE/DELETE/CREATE/INDEX/DROP/ALTER/CREATE TEMP TABLES/LOCK TABLES on matomo.* only; no FILE, no GRANT OPTION
TLS	Enabled, RDS CA bundle installed on EC2

Migration & validation

Fresh `mariadb-dump --single-transaction --routines --triggers` from EC2, restored directly into RDS. Row counts matched exactly before/after: `matomo_site` 1→1, `matomo_log_visit` 1→1. Config cutover (`config.ini.php [database]` section: host, TLS) confirmed working via HTTP 200 and the Matomo sign-in page rendering against RDS.

CRUD evidence (all via the running EC2 app, Administration -> Websites -> Manage)

Op	Action	Result
Create	Added website "Lab5-RDS-Test"	New row in matomo_site on RDS
Read	Viewed Manage Websites list	Site listed live from RDS
Update	Renamed to "Lab5-RDS-Test-Updated"	Change reflected immediately
Delete	Deleted the test site	Site count back to 1

Full screenshots for every step (instance running, RDS created, SG lockdown, all 4 CRUD ops) are in `Lab5-Screenshots.docx`.

Requirement checklist

☑ Deploy AWS RDS (MariaDB — justified deviation from "(MySQL/PostgreSQL)" wording, see above)

☑ Schema with 2+ tables (Matomo has 40+)

☑ Connect the Lab 4 EC2 app to RDS

☑ Demonstrate all CRUD from the running EC2 app

☑ RDS inbound restricted to the EC2 security group only, no 0.0.0.0/0